What Does Spiking Cost an Autoencoder? A Leakage-Controlled Comparison with a Matched Non-Spiking Control on Speech and EEG

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***Abstract*—Spiking autoencoders are usually compared with non-spiking networks that differ from them in more than the neuron, and mostly on images. We ask how much reconstruction quality the spike costs, and what it returns in arithmetic, for short speech and single-channel EEG windows. The spiking autoencoder (SNN-AE) is compared with a matched control (ANN-AE) that copies its selected widths, initializer and training and replaces only the leaky integrate-and-fire neuron by a leaky ReLU, and with PCA. The SNN-AE selects its architecture, the number of simulation steps, the initial threshold and decay, the spike encoding and an input transform by multi-objective search inside nested leave-one-group-out cross-validation on four datasets, and differences are tested at the recording level. Cost is an arithmetic energy estimate from the operations each forward pass executes. Searched LSTM, GRU and Transformer autoencoders, which differ from the SNN-AE in more than the neuron, were measured once, under one reduced search budget on development folds, for their training and inference cost and their quality, and left out of the main comparison. A design study on a development fold, excluded from the headline numbers, shows that two avoidable choices – scoring every simulation step and a spiking decoder – enlarged the gap, that the initializer decided the EEG comparison (with fan-in scaling in both arms the control was handicapped and the gap vanished), and that what remains is reproduced by a non-spiking network with a hard-clipped activation: the cost is the saturation of the neuron's transfer function, not the spike.**

***Index Terms*—spiking neural networks, autoencoder, matched control, energy estimate, nested cross-validation, speech, EEG**

I. Introduction

Spiking neural networks (SNNs) communicate with binary events, so a synapse whose input did not spike does no work, and a synapse whose input did spike adds a weight instead of multiplying by it [1], [2]. Surrogate gradients [3]–[5] made such networks trainable end to end, and autoencoders are an obvious use: compress a signal window to a small code on a low-power device and reconstruct it elsewhere.

Whether that trade is worth it is a question about two numbers measured on the *same* network: how much reconstruction error the spike adds, and how much arithmetic it removes. Published comparisons answer a different question. Spiking autoencoders have been reported as comparable to conventional ones on MNIST-type images, against baselines of similar size [6] or the same architecture [7]; when the comparison is against recurrent or attention models, those differ in recurrence, sequence length and capacity at once, so a quality difference cannot be attributed to the neuron. We are not aware of a matched comparison for the reconstruction of speech or EEG.

Two further confounds are easy to miss. First, evaluation: windows of one recording and one speaker are near-duplicates, so pooling windows before splitting measures memorization, and selecting hyperparameters on the test speaker biases every number [8]. Second, the comparison can be skewed by design choices that are not the spike, and in our case were: Section V documents a per-step loss and a spiking decoder that inflated the gap, and an initializer that, applied to both arms, handicapped the control enough to make the gap vanish on EEG.

This paper makes four contributions.

1) A *matched non-spiking control*: the ANN-AE copies the selected SNN-AE's widths, initializer, optimizer and data, and differs from it only in the encoder's hidden neuron, so the gap between them is the cost of spiking.
2) A leakage-controlled protocol: nested 6-fold leave-one-group-out cross-validation on two speech and two EEG datasets, with the SNN-AE's architecture selected on inner validation groups only, and recording-level paired statistics.
3) A quality–cost accounting in which the SNN's arithmetic is counted from the operations its forward pass executes (multiply–accumulates for analog inputs, additions for spikes), priced with one published per-operation energy table.
4) A design study, on a development fold kept out of the headline numbers, that locates where the SNN-AE's reconstruction error comes from, and a pre-validation on the same folds that measures what searched sequence autoencoders cost to train and to run.

II. Related Work

Spiking autoencoders. Comşa et al. [6] trained spiking autoencoders with temporal coding and found reconstruction errors comparable to conventional autoencoders of similar size on MNIST and Fashion-MNIST. The fully spiking VAE [7] compared against a non-spiking VAE of the same architecture on image datasets. Both support the matched design we adopt; neither addresses temporal signals or a group-disjoint protocol.

Training and coding. We train with surrogate gradients and backpropagation through time [3], [9], [10], and use the three

common input codes: direct currents, rate (Bernoulli) coding, and latency coding [11], [12].

Energy accounting. A widespread convention prices an SNN's analog-input layers by multiply–accumulates (MACs) and its spike-driven layers by accumulations (ACs), with 45 nm figures from Horowitz [13]; Spikformer [14] is one example. The same source shows that memory accesses cost one to three orders of magnitude more than the arithmetic, so such numbers rank architectures, they do not predict a chip's consumption.

Sequence autoencoders and protocol. LSTM [15], [16], GRU [17], [18] and self-attention [19], [20] autoencoders are the usual non-spiking baselines for sequences; because they differ from an SNN-AE in more than the neuron, we measure them in a pre-validation only (Section V-A). Group-disjoint, nested evaluation is the standard remedy for leakage in model selection [8].

III. Methods

A. Task

A window $x \in \mathbb{R}^M$ of $M = 256$ consecutive samples is z-scored on its own statistics and reconstructed through a latent $z \in \mathbb{R}^d$, minimizing the mean squared error. The latent width d is fixed per signal domain ($d = 16$ for speech, $d = 64$ for EEG) and is never searched, so every model faces the same compression ratio. A recording's last window is zero-padded; padded samples are excluded from normalization, every loss and every reported error.

B. SNN-AE

Neuron. The encoder's hidden layers are leaky integrate-and-fire (LIF) neurons [21] simulated for T steps:

$$V[t] = \beta\, V[t-1] + I[t], \qquad s[t] = \mathbf{1}[V[t] > V_{th}], \qquad (1)$$

and $V[t] \leftarrow 0$ after a spike (hard reset), with $\beta = e^{-\Delta t/(RC)}$ and $\Delta t = 1$. R, C and V_{th} are trained per layer (clamped above 10^{-6}) from initial values $R = 1$, $C = -1/\ln \beta^{(0)}$ and $V_{th}^{(0)}$, so the searched $\beta^{(0)}$ is the initial decay. The spike's zero gradient is replaced in the backward pass by the arctan surrogate [9] $\hat{\sigma}'(V) = \alpha/\big(2(1 + (\frac{\pi}{2}\alpha(V - V_{th}))^2)\big)$, $\alpha = 2$, and the error is propagated through time as

$$\delta_l[t] = \hat{\sigma}'(V_l[t]) \odot \big(W_{l+1}^{\top}\delta_{l+1}[t]\big) + \beta\, \delta_l[t+1]. \qquad (2)$$

Network. The encoder is $\mathrm{Lin}(M{\to}w_1) \to \mathrm{LIF} \to \cdots \to \mathrm{Lin}(w_{L-1}{\to}d)$ with $w_1 > \cdots > w_{L-1} > d$; the latent layer is analog. The latent is averaged over the T steps and decoded *once* by the mirror image of the encoder with leaky ReLUs (slope 0.01) and a linear output (Fig. 1). The loss is computed on that single reconstruction. Section V shows why: the earlier design, which scored the reconstruction of every step and spiked in the decoder too, was measurably worse for reasons unrelated to spiking.

Input. The window is presented for T steps as a constant current (*direct*), as Bernoulli spikes with probability proportional to the min–max-scaled sample (*Poisson*), or as one spike per sample at a time that decreases with its value (*latency*). An optional input transform is applied before the network: none (*dense*), a fixed 3-tap smoothing along the window (*conv1d*), or one LIF cell per sample with the network's threshold and decay (*recurrent*), which turns any code into spikes.

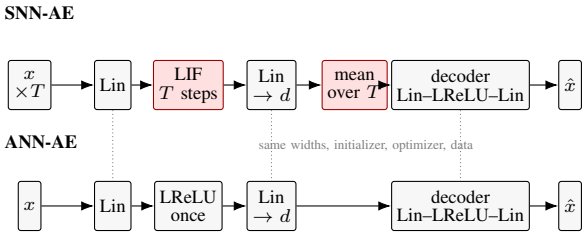


Figure 1. The SNN-AE and its matched control. Shared blocks have identical widths, initializer and training; only the encoder's hidden neuron differs (red: the parts that exist only in the spiking network).

C. Matched control (ANN-AE)

For every dataset, fold and run, the ANN-AE takes the widths of that run's selected SNN-AE and replaces each hidden LIF by a leaky ReLU applied once to the analog window; it is trained with the same initializer, optimizer, data and stopping rule. It is *not* searched. The SNN-AE therefore receives the larger search budget, which favours it, and a gap in the ANN-AE's favour is conservative.

D. References

PCA with $k = d$ components, fitted on all non-test windows of the fold, is the best linear code through the same bottleneck; a mean-window predictor fixes the scale of "no model".

E. Architecture search

The SNN-AE's architecture is selected by NSGA-II [22] with population 8 for 5 generations (48 evaluations), minimizing validation MSE and an analytic inference cost. The final front is re-scored over 3 seeds and the member with the lowest re-scored validation MSE wins (ties: lower cost). Genes: depth $L \in [1, 4]$ and widths in $(d, 128]$, $T \in [2, 32]$ (an integer drawn uniformly), $V_{th}^{(0)} \in [0.5, 1.5]$, $\beta^{(0)} \in [0.8, 0.99]$, encoding and input transform. The cost objective is a spike-blind MAC count in which T multiplies every encoder layer; it keeps cheaper genomes on the front, but the winner is chosen by validation MSE alone, so a longer simulation is selected only when it validates better.

F. Training

All models use Adam [23] (learning rate 10^{-3}; 10^{-4} for R, C, V_{th}), batch size 1, at most 30 epochs and early stopping with patience 5. The SNN-AE and the ANN-AE are both initialized with Xavier [24]. Left to their builders, the two arms had different defaults (fan-in scaling [25] for the SNN), and Section V shows that the initializer decides the EEG comparison, so it is fixed and shared. Within a fold the selected architecture is retrained on training ∪ validation groups, early-stopped on a recording-disjoint slice of the training groups, and evaluated once on the test group. Each fold is repeated 5 times (seeds 42–46), each repeat with its own search.

Table I
OPERATIONS COUNTED PER WINDOW (T STEPS, BINARY INPUT = SPIKES BY CONSTRUCTION).

Operation	Class	Count
Linear, binary input	AC	nonzero inputs × outputs
Linear, analog input	MAC	rows × inputs × outputs
LIF update, Eq. (1)	MAC	neurons × steps
Mean over T	AC, MAC	adds; one $1/T$ per output

Table II
DATASETS. G: GROUP UNIT; CAP: WINDOWS KEPT PER RECORDING; d: LATENT WIDTH.

Dataset	Signal	G	#G	cap	d
FSDD	speech, 8000 Hz	speaker	6	–	16
AudioMNIST	speech, 8000 Hz	speaker	60	2	16
EEGMMIDB	EEG, 160 Hz	subject	109	40	64
Siena	EEG, 512 Hz	subject	14	40	64

G. Cost

Cost is the arithmetic energy of one window's forward pass, $E = 4.6\,\mathrm{pJ} \cdot N_{\mathrm{MAC}} + 0.9\,\mathrm{pJ} \cdot N_{\mathrm{AC}}$, the 32-bit floating-point multiply–add and add of [13]. For the SNN-AE the counts come from the executed forward pass (Table I): a linear layer whose input is binary by construction (spikes, Poisson or latency input, the recurrent transform) costs one AC per nonzero input and output; any other linear layer costs MACs at every step it runs. The non-spiking models' MACs follow from their architectures (attention products included, Section V-A). Activations, biases and the input encoding are not counted for any model. E ranks architectures under one convention; it is not the energy of a chip, which memory traffic dominates [13].

IV. EXPERIMENTAL SETUP

Datasets (Table II). FSDD [26] and AudioMNIST are spoken digits at 8 kHz; EEGMMIDB [27], [28] (motor movement/imagery) and Siena [29] (epilepsy) are scalp EEG, read from the first channel of each recording (Fc5 and Fp1 in the file headers). The group is the speaker or subject. Windows do not overlap; long recordings are capped to a number of windows, and each fold keeps at most 1000 training, 150 validation and 1500 test windows, sampled round-robin across recordings.

Nested leave-one-group-out. Groups are sorted and split into 6 contiguous blocks. In fold f block f is the test set, block $f{+}1$ (mod 6) the validation set, and the rest train; for FSDD this is leave-one-speaker-out. Every window is assigned by its group before any pooling, and the harness aborts if a recording or group appears in two splits. All selection uses the validation block only.

Development fold. The design study (Section V) compared variants on fold 0 of FSDD and of EEGMMIDB, including their test windows. Those folds are therefore excluded from every headline number: FSDD and EEGMMIDB are reported on folds 1–5, AudioMNIST and Siena on folds 0–5. Including them is reported as a sensitivity analysis (Table VIII).

Table III
DESIGN STUDY, DEVELOPMENT FOLD, TEST MSE (MEAN OF 3 SEEDS; −: NOT RUN). KAIMING / XAVIER: THE INITIALIZER OF EVERY VARIANT IN THE COLUMN.

Variant	FSDD		EEGMMIDB	
	Kaiming	Xavier	Kaiming	Xavier
ANN, leaky ReLU (control)	0.667	0.654	0.489	0.293
ANN, sigmoid	0.685	–	0.308	–
ANN, clamp to $[0, 1]$	0.743	–	0.547	–
SNN, per-step loss, LIF dec., T=16	0.780	–	0.592	0.589
SNN, loss on time mean, T=16	0.734	–	0.454	–
SNN, latent mean, LReLU dec., T=4	0.724	0.729	0.484	0.449
SNN, latent mean, LReLU dec., T=16	0.729	–	0.457	0.396

Statistics. Every model reconstructs the same test windows. The primary estimand is the recording-level paired difference $d_r = E_{m,r} - E_{b,r}$, the SNN-AE's mean window error on recording r minus a reference model's: we report $\overline{d_r}$, a 95% bootstrap interval over recordings, and the Wilcoxon signed-rank p, Holm-corrected across the three references [30]. Group-level and seed-level tests are reported as robustness only; window-level statistics are pseudoreplicated and descriptive.

V. DESIGN STUDY ON THE DEVELOPMENT FOLD

Before the main run we asked where the gap between an earlier SNN-AE design and the matched control came from, varying one factor at a time on the development fold (1000 training windows, the search's most frequent winner, three seeds). The variants were implemented in a PyTorch replica of the network, validated against the C++ harness (EEGMMIDB: earlier SNN-AE 0.592 vs. 0.593; FSDD: ANN-AE 0.667 vs. 0.662). Table III lists the test MSE.

Three findings shaped the method. *(i) Two costs were not the spike.* Scoring the reconstruction of every simulation step asks each step, carrying at most one binary event per neuron, to reproduce the analog window alone; scoring the time mean removed 0.046 on FSDD. A spiking decoder added error on top of the encoder's; averaging the latent and decoding once with leaky ReLUs is as good and runs the decoder once instead of T times. *(ii) The initializer decided the EEG comparison.* With fan-in scaling in both arms the control stopped at 0.489 (0.479 with a 200-epoch cap, so not for lack of epochs) and the SNN-AE looked equal to it (0.484); with Xavier in both arms the control reaches 0.293 and the SNN-AE 0.449. Xavier is the better or equal choice for each arm on both datasets, so the matched comparison is also a best-against-best one. *(iii) What remains is saturation.* A non-spiking network with a hard clamp lands among the spiking variants (FSDD 0.743, against 0.734 for the SNN trained on the time mean; EEG 0.547, against 0.592 for the per-step SNN), while a sigmoid, bounded but smooth, stays near the control (FSDD 0.685 vs. 0.667; EEG 0.308, near the Xavier control's 0.293). A reason, argued rather than measured: under direct input an LIF neuron sees the same current I at every step, so with a hard reset it never fires for $I < V_{th}(1 - \beta)$ and fires at every step for $I \geq V_{th}$ – a clipped transfer function.

T mattered on EEG (0.449 at T=4, 0.396 at T=16) but not on FSDD (0.717 to 0.729 for T from 1 to 16), which is why T is searched rather than fixed. Every choice above came out the same on the fold's validation windows (e.g. EEG, Xavier: 0.425 vs. 0.372 for $T = 4$ vs. 16), but the test windows were seen, hence the development-fold rule of Section IV. A check of the adopted design in the C++ harness on the same fold (two runs, small search) moved the SNN-AE from 0.827 to 0.768 on FSDD and from 0.593 to 0.532 on EEG, against controls of 0.652 and 0.302.

A. Pre-validation of sequence baselines

The submitted version also searched LSTM, GRU and Transformer autoencoders. They differ from the SNN-AE in recurrence, sequence length and capacity at once, so they cannot isolate what the spike costs, and their searches dominated the grid's compute. Before leaving them out we measured them once, on the development folds (FSDD fold 0 and EEGMMIDB fold 0). LSTM-AE and GRU-AE stack $L \in [1, 3]$ recurrent layers of width $H \in [8, 256]$ (floored at d), project the last state to d and decode by repeating a projection of z over time. The Transformer-AE embeds frames to $d_{\text{model}} \in [16, 128]$, adds sinusoidal positions, applies $N \in [1, 4]$ self-attention blocks (heads $\in \{1, 2, 4, 8\}$, feed-forward width $\in [32, 256]$), mean-pools to d and decodes with N further blocks and no cross-attention, so all information passes through z. All three read the window as $S = 32$ frames of 8 samples, so only the direct code is defined for them. Every family, the SNN-AE included, ran one NSGA-II of population 4 for 2 generations (12 evaluations, front re-scored over 2 seeds), 2 runs (seeds 42–43), with the windows, training and selection of Sections III-E–III-F, each family in its own process on one CPU thread.

Table IV reports the cost of training (the fold's wall time) and of processing data after training (the measured time to reconstruct one window, and E): each of the three families' runs took at least 1.7 times the SNN-AE's wall time on the same fold, each winner took at least 2.7 times the SNN-AE's time to reconstruct a window, and each winner needed at least 4.5 times the SNN-AE's arithmetic energy per window. On quality, none of the three reached the test error of PCA on either development fold. The reduced budget can change any family's winner, and with it both its error and its per-window cost; the table describes the winners found here, not the families' best.

Table IV
PRE-VALIDATION, DEVELOPMENT FOLDS, 2 RUNS. FOLD: WALL TIME OF THE FAMILY'S FOLD, DATA LOADING EXCLUDED (SEARCH, RE-SCORING, FINAL FIT, TEST); ×SNN: ITS RATIO TO THE SNN-AE'S, WHOSE FOLD ALSO TRAINS THE ANN-AE. TRAIN: FINAL FIT, PER TRAINING WINDOW AND EPOCH. INFER.: ONE WINDOW RECONSTRUCTED AFTER TRAINING. ONE CPU THREAD EACH.

Model	Test MSE	Fold (h)	×SNN	Train (ms/win.)	Infer. (ms/win.)	E (nJ)
FSDD						
SNN-AE	0.712 ± 0.028	0.52	1.0	1.51	0.27	1081
ANN-AE	0.668 ± 0.000	–	–	0.97	0.08	118
LSTM-AE	0.888 ± 0.010	3.58	6.9	8.17	1.99	4894
GRU-AE	0.929 ± 0.095	5.01	9.6	35.0	10.5	30510
Transformer-AE	0.998 ± 0.000	1.00	1.9	4.42	0.73	10678
PCA ($k = d$)	0.649	–	–	–	–	–
EEGMMIDB						
SNN-AE	0.416 ± 0.016	0.60	1.0	1.67	0.39	302
ANN-AE	0.294 ± 0.002	–	–	1.02	0.08	191
LSTM-AE	1.000 ± 0.000	4.83	8.0	32.1	8.58	35878
GRU-AE	0.641 ± 0.003	9.70	16	65.1	17.9	68990
Transformer-AE	0.997 ± 0.005	1.04	1.7	6.28	1.04	17265
PCA ($k = d$)	0.183	–	–	–	–	–

VI. RESULTS

Draft: the tables of this section are placeholders (−) until the nested-LOSO grid has run; they are regenerated from the harness's outputs.

Table V gives the test MSE and energy of every model on the headline folds, Table VI the recording-level differences between the SNN-AE and each reference, Fig. 2 the same models in the quality–energy plane, and Table VII what the SNN-AE's search selected.

Reading guide. The first question is the gap between the SNN-AE and its matched control, per dataset and modality, and whether its interval excludes zero. The second is whether the SNN-AE buys arithmetic for it: its E against the ANN-AE's, read together with the selected encoding and T in Table VII, since binary inputs turn the first layer's MACs into ACs while direct input pays a dense first layer at every step. The third places both against PCA; a model that does not beat PCA has not shown that its nonlinearity helps on that dataset. Orderings are asserted only where the Holm-corrected p is below 0.05 and the interval excludes zero, and a finding that holds on one modality only is reported as such.

VII. DISCUSSION

The design study already constrains what the main run can show. The spike's cost in this autoencoder is not a property of discrete events as such: a non-spiking network with the same clipped transfer function pays it, and a smooth bounded one does not. Remedies that act on the transfer function – a membrane read out instead of spikes in some layers, adaptive thresholds, or soft resets, which made things worse here – are therefore the levers to test. Longer simulations helped on EEG only (Table III) and cost arithmetic in proportion to T. On cost, the convention we follow rewards binary inputs: whether the SNN-AE ends up cheaper than its control depends on which encoding the search selects, not on spiking alone.

Limitations. (i) *Groups.* Recordings are nested in 6, 60, 109 and 14 groups (5 FSDD speakers on the headline folds). The recording-level test treats recordings as independent, so it speaks to new recordings of these speakers and subjects, not to new ones; the group-level tests that would are reported as robustness only, and with 5 speakers the exact Wilcoxon p cannot fall below 0.0625. (ii) *One channel, short windows.* EEG is read from one channel and every window has 256 samples; results may not transfer to multichannel or longer contexts. (iii) *Estimated, not measured energy.* E counts arithmetic only; memory traffic, sparsity exploitation and the target hardware decide real consumption. (iv) *Search budget.* The SNN-AE evaluates 48 candidates per fold and run, and the control none

Table V

TEST MSE (MEAN ± STD OVER 5 SEEDS; LOWEST PER DATASET IN BOLD) AND ARITHMETIC ENERGY PER WINDOW E (NJ, SECTION III-G) ON THE HEADLINE FOLDS. PCA AND MEAN ARE FITTED ONCE (NO SEEDS, NO STD) AND ARE NOT COSTED.

Model	FSDD MSE	E	AudioMNIST MSE	E	EEGMMIDB MSE	E	Siena MSE	E
Mean	–	–	–	–	–	–	–	–
PCA	–	–	–	–	–	–	–	–
SNN-AE	–	–	–	–	–	–	–	–
ANN-AE (matched)	–	–	–	–	–	–	–	–

Table VI

SNN-AE MINUS REFERENCE, RECORDING LEVEL, HEADLINE FOLDS: $\overline{d_r}$ (POSITIVE: THE SNN-AE ERRS MORE; *: HOLM-CORRECTED WILCOXON $p < 0.05$) AND ITS 95% BOOTSTRAP INTERVAL.

Dataset	n_{rec}	ANN-AE	PCA	Mean
FSDD	–	–	–	–
AudioMNIST	–	–	–	–
EEGMMIDB	–	–	–	–
Siena	–	–	–	–

Quality–energy plane: generated from data/paper_loso_⟨dataset⟩_pareto.csv after the grid run.

Figure 2. Test MSE (mean ± std over seeds) against arithmetic energy per window, headline folds. Down and left is better; the dashed line is PCA with $k = d$.

Table VII

SNN-AE SELECTED PER DATASET OVER EVERY (FOLD, RUN): MODAL ENCODING AND INPUT TRANSFORM (k/n AGREEING, IF NOT ALL), MEDIAN T [RANGE], MEDIAN INITIAL THRESHOLD AND DECAY, MEDIAN DEPTH L, AND THE HIDDEN NEURONS' SPIKE FRACTION ρ ON TEST WINDOWS.

Dataset	Enc.	Transf.	T	$V_{th}^{(0)}$	$\beta^{(0)}$	L	ρ
FSDD	–	–	–	–	–	–	–
AudioMNIST	–	–	–	–	–	–	–
EEGMMIDB	–	–	–	–	–	–	–
Siena	–	–	–	–	–	–	–

Table VIII

SENSITIVITY TO THE DEVELOPMENT FOLD: SNN-AE MINUS ANN-AE, RECORDING LEVEL, WITHOUT (HEADLINE) AND WITH IT.

Dataset	Folds	n_{rec}	$\overline{d_r}$	95% CI	p_{Holm}
FSDD	–	–	–	–	–
	–	–	–	–	–
EEGMMIDB	–	–	–	–	–
	–	–	–	–	–

(it inherits the SNN-AE's widths), which favours the SNN-AE over its control; a larger budget could change the winner. (v) *Matched, not identical inputs.* The ANN-AE always reads the analog window, whereas the SNN-AE may read spikes; the information available is the same, its code is not. (vi) *Fixed latent width.* d follows prior autoencoder practice for each domain, not a sweep on these datasets. (vii) *Training cap.* At most 30 epochs; on the development fold most EEG runs were still improving at the cap. (viii) *Sequence models pre-validated, not compared.* The LSTM-, GRU- and Transformer-AE were searched with 12 evaluations on two development folds only, so the conclusions concern spiking against its matched control, not against sequence models.

VIII. CONCLUSION

We compared a spiking autoencoder with a matched non-spiking control and PCA on two speech and two EEG datasets, under nested leave-one-group-out cross-validation with every choice made on validation groups and the development fold kept out of the headline numbers; searched sequence autoencoders were measured in a pre-validation on the same folds for what they cost to train and to run. Matching the control exposed two confounds that would have changed the conclusion – a per-step loss and a spiking decoder that inflated the gap, and an initializer that, shared but poorly chosen, made it vanish on EEG – and located the remaining cost in the saturation of the

LIF transfer function. The harness, profile and analysis scripts are released [31].

REFERENCES

[1] W. Maass, "Networks of spiking neurons: The third generation of neural networks," *Neural Networks*, vol. 10, no. 9, pp. 1659–1671, 1997.

[2] M. Pfeiffer and T. Pfeil, "Deep learning with spiking neurons: Opportunities and challenges," *Frontiers in Neuroscience*, vol. 12, p. 774, 2018.

[3] E. O. Neftci, H. Mostafa, and F. Zenke, "Surrogate gradient learning in spiking neural networks," *IEEE Signal Processing Magazine*, vol. 36, no. 6, pp. 51–63, 2019.

[4] F. Zenke and S. Ganguli, "Superspike: Supervised learning in multilayer spiking neural networks," *Neural Computation*, vol. 30, no. 6, pp. 1514–1541, 2018.

[5] F. Zenke and T. P. Vogels, "The remarkable robustness of surrogate gradient learning for instilling complex function in spiking neural networks," *Neural Computation*, vol. 33, no. 4, pp. 899–925, 2021.

[6] I.-M. Comşa, L. Versari, T. Fischbacher, and J. Alakuijala, "Spiking autoencoders with temporal coding," *Frontiers in Neuroscience*, vol. 15, p. 712667, 2021.

[7] H. Kamata, Y. Mukuta, and T. Harada, "Fully spiking variational autoencoder," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 6, 2022, pp. 7059–7067.

[8] G. C. Cawley and N. L. C. Talbot, "On over-fitting in model selection and subsequent selection bias in performance evaluation," *Journal of Machine Learning Research*, vol. 11, pp. 2079–2107, 2010.

[9] J. K. Eshraghian *et al.*, "Training spiking neural networks using lessons from deep learning," *Proceedings of the IEEE*, vol. 111, no. 9, pp. 1016–1054, 2023.

[10] P. J. Werbos, "Backpropagation through time: What it does and how to do it," *Proceedings of the IEEE*, vol. 78, no. 10, pp. 1550–1560, 1990.

[11] S. Thorpe, A. Delorme, and R. Van Rullen, "Spike-based strategies for rapid processing," *Neural Networks*, vol. 14, no. 6–7, pp. 715–725, 2001.

[12] I. M. Comsa, K. Potempa, L. Versari, T. Fischbacher, A. Gesmundo, and J. Alakuijala, "Temporal coding in spiking neural networks with alpha synaptic function," in *ICASSP 2020 – 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2020, pp. 8529–8533.

[13] M. Horowitz, "Computing's energy problem (and what we can do about it)," in *IEEE International Solid-State Circuits Conference (ISSCC) Digest of Technical Papers*, 2014, pp. 10–14.

[14] Z. Zhou, Y. Zhu, C. He, Y. Wang, S. Yan, Y. Tian, and L. Yuan, "Spikformer: When spiking neural network meets transformer," in *International Conference on Learning Representations (ICLR)*, 2023.

[15] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[16] I. Sutskever, O. Vinyals, and Q. V. Le, "Sequence to sequence learning with neural networks," in *Advances in Neural Information Processing Systems*, vol. 27, 2014, pp. 3104–3112.

[17] K. Cho, B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, "Learning phrase representations using RNN encoder–decoder for statistical machine translation," in *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2014, pp. 1724–1734.

[18] J. Chung, C. Gulcehre, K. Cho, and Y. Bengio, "Empirical evaluation of gated recurrent neural networks on sequence modeling," *arXiv preprint arXiv:1412.3555*, 2014.

[19] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 5998–6008.

[20] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," *arXiv preprint arXiv:1607.06450*, 2016.

[21] W. Gerstner and W. M. Kistler, *Spiking Neuron Models: Single Neurons, Populations, Plasticity*. Cambridge University Press, 2002.

[22] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan, "A fast and elitist multiobjective genetic algorithm: NSGA-II," *IEEE Transactions on Evolutionary Computation*, vol. 6, no. 2, pp. 182–197, 2002.

[23] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *International Conference on Learning Representations (ICLR)*, 2015.

[24] X. Glorot and Y. Bengio, "Understanding the difficulty of training deep feedforward neural networks," in *Proceedings of the 13th International Conference on Artificial Intelligence and Statistics (AISTATS)*, ser. PMLR, vol. 9, 2010, pp. 249–256.

[25] K. He, X. Zhang, S. Ren, and J. Sun, "Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2015, pp. 1026–1034.

[26] Z. Jackson *et al.*, "Free spoken digit dataset," GitHub repository, 2018. [Online]. Available: https://github.com/Jakobovski/free-spoken-digit-dataset

[27] G. Schalk, D. J. McFarland, T. Hinterberger, N. Birbaumer, and J. R. Wolpaw, "BCI2000: A general-purpose brain-computer interface (BCI) system," *IEEE Transactions on Biomedical Engineering*, vol. 51, no. 6, pp. 1034–1043, 2004.

[28] A. L. Goldberger, L. A. N. Amaral, L. Glass, J. M. Hausdorff, P. C. Ivanov, R. G. Mark, J. E. Mietus, G. B. Moody, C.-K. Peng, and H. E. Stanley, "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000.

[29] P. Detti, G. Vatti, and G. Zabalo Manrique de Lara, "EEG synchronization analysis for seizure prediction: A study on data of noninvasive recordings," *Processes*, vol. 8, no. 7, p. 846, 2020.

[30] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.

[31] Anonymous, "Reference suppressed for double-blind review," 2026, code repository citation withheld; to be restored in the camera-ready version.